

Xinglang Zhang

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Education

Huazhong University of Science and Technology (HUST)

Sept 2024 – Jun 2028

B.S. in Cyber Science and Engineering

- **GPA:** 4.79/5.0 **Rank:** 1/108
- **Selected Coursework:** Calculus A (I/II) (A+), Linear Algebra (A+), Discrete Mathematics (I/II) (A+), Data Structures (A+), Assembly Language (A+), Probability Theory (A+), Physics A (A+), Database Systems (A+).
- **Research Interests:** LLM Reasoning, Logical Reasoning, Neural-Symbolic AI, AI Safety

Publications

Logical Phase Transitions: Understanding Collapse in LLM Logical Reasoning

Xinglang Zhang*, Yunyao Zhang*, et al. | [\[ArXiv\]](#) [\[Code\]](#)

From Ambiguity to Verdict: A Semiotic-Grounded Multi-Perspective Agent for LLM Logical Reasoning

Yunyao Zhang, Xinglang Zhang, et al. | [\[ArXiv\]](#) [\[Code\]](#)

Research Experience

Exploring Logical Reasoning Boundaries and Enhancement Strategies in LLMs via Neural-Symbolic Methods

Aug 2025 – Jan 2026

Project Leader, Intelligent Media Computing and Network Security Lab, HUST

Advisor: [Zikai Song](#), Research Areas: LLM Evaluation, Logical Reasoning, Neural-Symbolic AI

- **Framework:** Identified *Logical Phase Transitions* in LLM reasoning and proposed a *Neural-Symbolic Tuning* framework to stabilize reasoning under increasing logical complexity.
- **Results:** Across five benchmarks, the method mitigates reasoning collapse, yielding +1.26 (naive) and +3.95 (CoT) average accuracy gains with improved generalization.
- **Outcome:** The work was submitted as first author to **ACL 2026**.

Reasoning Framework for Large LLMs based on Philosophical Paradigms

Feb 2025 – Aug 2025

Research Assistant, Intelligent Media Computing and Network Security Lab, HUST

Advisor: [Zikai Song](#)

Research Areas: LLM Reasoning, Neural-Symbolic AI, Semantic and Symbolic Reasoning

- **Framework & Benchmark:** Proposed *LogicAgent*, a semiotic-aware logical reasoning framework, and introduced *RepublicQA*, a benchmark jointly modeling logical and semantic complexity.
- **Results:** Achieved state-of-the-art results on RepublicQA (+6.25%) and strong generalization across four logical reasoning benchmarks (+7.05%).
- **Outcome:** Co-authored a paper submitted to **ACL 2026**.

Safety-Aligned Embodied AI: A Comprehensive Survey & Analysis

Aug 2025 – Oct 2025

Research Assistant, AIM3 Lab, Renmin University of China (RUC)

Advisor: [Wenxuan Wang](#), Research Areas: Embodied AI, AI Safety

- **Framework:** Reviewed safety-aligned Embodied AI research, organizing representative methods across perception, planning, and action.
- **Progress:** Analyzed over 30 papers to summarize common experimental settings, evaluation protocols, and safety considerations.
- **Outcome:** Contributed to literature synthesis and early-stage structuring of an in-progress survey project.

Honors

- **National Scholarship** (Top 3.2%) 2024 – 2025
- **Triple-A ("Sanhao") Student Scholarship** (Top 5.5%) 2024 – 2025
- **Outstanding Undergraduate Student**, Qiming College (Top 1.85%) 2025 – 2027
- **Outstanding Communist Youth League Member** (Top 4.1%) 2024 – 2025
- **Academic Excellence Scholarship & Self-improvement Scholarship** 2024 – 2025

Skills

ML & Research Stack: Python, C/C++, PyTorch, HuggingFace, LoRA, SFT, RLHF, vLLM

Mathematical Foundations: Linear Algebra, Probability Theory, Calculus, Discrete Mathematics

Languages: English (CET-4: 609; CET-6: pending), Chinese (Native)